

Mayank Soni

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Education

University of Hyderabad

2023 – Expected 2028

Integrated B.Sc. + M.Sc. in Physics

Research Interests

Topological Physics • Condensed Matter Theory • Singular Optics • Computational Physics

Research Experience

Undergraduate Researcher

June 2025 – Present

University of Hyderabad

Advisor: Prof. Nirmal K. Viswanathan

- Investigated three-dimensional polarization singularities in electromagnetic vector fields.
- Developed computational methods for simulation and analysis of optical speckle fields.
- Created numerical tools for visualization and tracking of optical singular structures.
- Contributed to the preparation of a manuscript on polarization singularities and optical speckle fields.

Summer Research Intern

May 2024 – August 2024

Krittika Summer Projects, IIT Bombay

- Analyzed observational data from SN 2023ixf using Python.
- Constructed multi-band light curves from astronomical observations.
- Estimated physical parameters of the supernova through numerical analysis.

Scientific Software

VectorWaves

December 2025 – June 2026

- Developed and maintain an open-source Python package for constructing and analyzing electromagnetic fields using plane-wave expansions.
- Implemented GPU-accelerated field evaluation, visualization, and singularity-analysis workflows.
- Published the package on [PyPI](https://pypi.org/project/VectorWaves/), with source code on [GitHub](https://github.com/1Rayokelvin/VectorWaves) with documentation [here](#).

Presentations & Manuscripts

- **Oral Presentation:** “Three-Dimensional Polarization Structure: From Paraxial to Full-Vector Singularities,” SCOPOSIS-2025, Physical Research Laboratory, Ahmedabad.
- **Poster Presentation:** “Three-Dimensional Analysis of Polarisation Singularities, Paraxial to Isotropic Regimes,” 100 Years of Quantum Mechanics, University of Hyderabad.
- **Manuscript in Preparation:** “Dimensional Crossover in Gaussian Speckles and its Topological Consequences” (with Prof. Nirmal K. Viswanathan).

Technical Skills

Programming: Python, Julia

Scientific Computing: NumPy, SciPy, Matplotlib

Tools: Git, LaTeX